Geometric Deep Learning Notes

UW-Madison, The RTG Mathematics and AI Synergies (MAIS) GDL reading group

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1 From Geometric priors to NN architecture

1.1 Background

The input we consider is not a feature vector in $\mathbb{R}^n$ anymore, but a signal on a space Ω.

Definition 1.1 (Signal)**.** *A signal on a domain Ω is a function*

$$x : \Omega \to C,$$

where C is a vector space. Its dimension is called the number of channels.

For example, for an image, Ω is a 2D grid and the signal is an RGB value. For a graph, Ω is the set of nodes and the signal is a feature assigned to each node.

Consider the space of C-valued signals on Ω:

$$\mathcal{X}(\Omega, C) = \{x : \Omega \to C\}.$$

This is a real vector space with pointwise addition and scalar multiplication:

$$(\alpha x + \beta y)(u) = \alpha x(u) + \beta y(u), \qquad \forall u \in \Omega, \quad \alpha, \beta \in \mathbb{R}.$$

Given an inner product $\langle v, w \rangle_C$ on C and a measure μ on Ω, we can define an inner product on $\mathcal{X}(\Omega, C)$ by

$$\langle x, y \rangle = \int_\Omega \langle x(u), y(u) \rangle_C \, d\mu(u).$$

Here $u \mapsto \langle x(u), y(u) \rangle_C$ is a function $\Omega \to \mathbb{R}$ (or $\Omega \to \mathbb{C}$).

1.2 Groups and Symmetry

1.2.1 Group action

Recall that a left group action is a map $G \times \Omega \to \Omega$, denoted by $g \cdot u$, satisfying

$$e \cdot u = u, \qquad g \cdot (h \cdot u) = (gh) \cdot u.$$

The orbit of an element u and its stabilizer are

$$G \cdot u = \{g \cdot u \mid g \in G\}, \qquad \mathrm{Stab}_G(u) = \{g \in G \mid g \cdot u = u\}.$$

An action is called transitive if there is only one orbit. It is called free if

$$\mathrm{Stab}_G(u) = \{e\} \qquad \forall u \in \Omega.$$

An action is called *faithful* if the only group element that fixes every point is the identity. More precisely, define the kernel of the action by

$$\ker(G \curvearrowright \Omega) := \{g \in G : g \cdot u = u \text{ for every } u \in \Omega\} = \bigcap_{u \in \Omega} \mathrm{Stab}_G(u).$$

The action is faithful if and only if

$$\ker(G \curvearrowright \Omega) = \{e\}.$$

Equivalently, for every $g \neq e$, there is some $u \in \Omega$ such that $g \cdot u \neq u$. If we have $G \curvearrowright \Omega$, we automatically have an action on signals:

$$G \times \mathcal{X}(\Omega, C) \to \mathcal{X}(\Omega, C), \qquad (g, x) \mapsto g \cdot x,$$

where $g \cdot x : \Omega \to C$ is given by

$$(g \cdot x)(u) = x(g^{-1} \cdot u).$$

Indeed,

$$(e \cdot x)(u) = x(e^{-1} \cdot u) = x(u),$$

and

$$\begin{aligned}(g \cdot (h \cdot x))(u) &= (h \cdot x)(g^{-1} \cdot u) \\ &= x\big(h^{-1} \cdot (g^{-1} \cdot u)\big) \\ &= x\big((gh)^{-1} \cdot u\big) \\ &= ((gh) \cdot x)(u).\end{aligned}$$

Thus $g \cdot (h \cdot x) = (gh) \cdot x$.

Now define the symmetric group of Ω:

$$\mathrm{Sym}(\Omega) = \{\varphi : \Omega \to \Omega \mid \varphi \text{ is bijective}\}.$$

For each fixed $g \in G$, the action map

$$\varphi_g : \Omega \to \Omega, \qquad u \mapsto g \cdot u,$$

is a bijection. We can also add structures and demand that φ_g preserve them. For example:

Structure on Ω	Condition on φ_g	Symmetry group
General set	φ_g is bijective	$\mathrm{Sym}(\Omega)$
Graph	$(u,v) \in E \iff (g \cdot u, g \cdot v) \in E$	$\mathrm{Aut}(\Gamma)$
Riemannian manifold	$\varphi_g^* h = h$	$\mathrm{Isom}(M)$
Vector space	φ_g is linear	$\mathrm{GL}(V)$

Thus, a group action induces a representation.

1.2.2 Group representations

Recall that a representation of a group G on V is a homomorphism

$$\rho : G \to \mathrm{GL}(V)$$

satisfying

$$\rho(gh) = \rho(g)\rho(h), \qquad \rho(e) = I.$$

If we choose a basis $\mathcal{B}$ of V, we get a matrix representation under the basis $\mathcal{B}$:

$$\rho_{\mathcal{B}} : G \to \mathrm{GL}(n, \mathbb{R}).$$

A subrepresentation is a subspace $W \leq V$ such that $\rho(g)W \subseteq W$ for every $g \in G$. Then W is called an invariant subspace, and the restriction of ρ to W is a subrepresentation. A representation is irreducible if it has no invariant subspace except $\{0\}$ and V.

An intertwiner is a linear map $A : V \to V'$ such that

$$A\rho(g) = \rho'(g)A.$$

In particular, an n-dimensional real representation of G is a map

$$\rho : G \to \mathbb{R}^{n \times n},$$

where $\rho(g)$ is an invertible matrix and

$$\rho(gh) = \rho(g)\rho(h).$$

A representation is called unitary or orthogonal if $\rho(g)$ is unitary or orthogonal for every $g \in G$.

Using the language of representation, the action $G \curvearrowright \mathcal{X}(\Omega, C)$ can be written as

$$\rho(g) : \mathcal{X}(\Omega, C) \to \mathcal{X}(\Omega, C), \qquad x \mapsto g \cdot x,$$

with

$$[\rho(g)x](u) = x(g^{-1} \cdot u).$$

1.2.3 Invariant and equivariant functions

Definition 1.2 (G-invariant function)**.** *A function $f : \mathcal{X}(\Omega, C) \to Y$ is G-invariant if*

$$f(\rho(g)x) = f(x), \qquad \forall g \in G,\ x \in \mathcal{X}(\Omega, C).$$

That is, the output is unaffected by the group action on the input.

Definition 1.3 (G-equivariant function)**.** *A function $f : \mathcal{X}(\Omega, C) \to \mathcal{X}(\Omega, C')$ is G-equivariant if*

$$f(\rho(g)x) = \rho(g)f(x), \qquad \forall g \in G,\ x \in \mathcal{X}(\Omega, C).$$

For an invariant function, the output may be a classification or a probability in Y. For an equivariant function, the input and output transform together according to the group action.

1.2.4 Example: image classification and CNNs

For an image, let $\Omega = \mathbb{Z}^2$ be the set of pixels (with no boundary), and let $u = (i, j) \in \Omega$ be the position of a pixel. A signal

$$x : \Omega \to \mathbb{R}^3$$

is the color channel data. We can regard x as a specific picture, and $x(u)$ is the pixel value at u. Consider the shift group (2D)

$$G = (\mathbb{Z}^2, +).$$

For any $a = (a_1, a_2) \in G$, let it act on Ω by

$$a \cdot u = u + a.$$

It is an isometry under the Euclidean metric $g_{ij} = \delta_{ij}$. Now consider $G \curvearrowright \mathcal{X}(\Omega, \mathbb{C})$. Its induced action is

$$[\rho(a)x](u) = x(a^{-1} \cdot u) = x(u - a).$$

Thus, when the group element a acts on a pixel, its new position is $u + a$. If f is the CNN classifier,

$$f : \mathcal{X}(\Omega) \to [0, 1] \quad \text{or} \quad f : \mathcal{X}(\Omega) \to \{0, 1\},$$

then it is G-invariant by the architecture of convolution. If f is a CNN layer,

$$f : \mathcal{X}(\Omega, C) \to \mathcal{X}(\Omega, C'),$$

then it is G-equivariant. We show this for K output channels and three input channels. Let θ_{kc} be the filters, for $k = 1, \ldots, K$ and $c = 1, 2, 3$, and define

$$y_k(i, j) := (x * \theta)_k(i, j) = \sum_{c=1}^{3} \sum_{(a,b) \in \mathbb{Z}^2} \theta_{kc}(i - a, j - b) x_c(a, b),$$

where each filter has finite support (for example, $\operatorname{supp} \theta_{kc} \subseteq \{-1, 0, 1\}^2$). Define T_t to be translation by $t = (p, q) \in \mathbb{Z}^2$:

$$(T_t x)_c(i, j) = x_c(i - p, j - q).$$

We claim that

$$(T_t x) * \theta = T_t(x * \theta), \qquad \forall t = (p, q) \in \mathbb{Z}^2.$$

Indeed, for each output channel k,

$$\begin{aligned}((T_t x) * \theta)_k(i,j) &= \sum_{c=1}^{3} \sum_{(a,b)\in\mathbb{Z}^2} \theta_{kc}(i-a, j-b)(T_t x)_c(a,b) \\ &= \sum_{c=1}^{3} \sum_{(a,b)\in\mathbb{Z}^2} \theta_{kc}(i-a, j-b) x_c(a-p, b-q) \\ &= \sum_{c=1}^{3} \sum_{(a',b')\in\mathbb{Z}^2} \theta_{kc}((i-p)-a', (j-q)-b') x_c(a',b') \\ &= (x * \theta)_k(i-p, j-q) \\ &= [T_t(x*\theta)]_k(i,j),\end{aligned}$$

where $a' = a - p$ and $b' = b - q$. Thus the two sides are equal for every (i,j) and every output channel k.

1.3 Metric Geometry and Deformation Stability

- The domain Ω is fixed and the signal undergoes deformation.
- The domain Ω itself undergoes deformation.

1.3.1 Stability of Signals on a Fixed Domain

Group actions such as translations describe global symmetries of Ω. Shifting an image of a cat should preserve its label. What if the cat turns its head to the left while its body stays in place? Small deformations generally fail to form a group because their composition can be large. Below, Ω is a metric space with d

Definition 1.4 (Complexity measure)**.** *Let*

$$\tau \in \mathrm{Sym}(\Omega), \qquad G \le \mathrm{Isom}(\Omega, d) \le \mathrm{Sym}(\Omega).$$

The complexity measure $c(\tau)$ measures "how far" τ is from G, with $c(\tau) = 0$ for $\tau \in G$.

Definition 1.5 (geometric stable)**.** *A function $f \in \mathcal{F}(\mathcal{X}(\Omega))$ is called geometrically stable if, for a constant C,*

$$\|f(\rho(\tau)x) - f(x)\| \le C c(\tau) \|x\|_{\mathcal{X}(\Omega)} \qquad \text{for all } x \in \mathcal{X}(\Omega) \text{ and } \tau \in \mathrm{Sym}(\Omega).$$

This is a form of approximate invariance.

1.3.2 Stability to Domain Deformation

Definition 1.6 (Domain distance)**.** *Let $\mathcal{D}$ be a class of geometric domains, for instance graphs or Riemannian manifolds. We want a distance on $\mathcal{D}$ that vanishes for isomorphic graphs and for isometric Riemannian manifolds. For metrics d and $\widetilde{d}$ on Ω and $\widetilde{\Omega}$, respectively, consider*

$$d_{\mathcal{D}}(\Omega, \widetilde{\Omega}) = \inf_{\eta} \left\| d - \widetilde{d} \circ (\eta \times \eta) \right\|,$$

where the infimum is over bijections $\eta : \Omega \to \widetilde{\Omega}$ (or admissible bijections when additional structure is imposed, such as isometry).

The norm acts on real-valued functions on $\Omega \times \Omega$. In particular,

$$\Delta_\eta(u,v) = d(u,v) - \widetilde{d}(\eta(u),\eta(v)), \quad \|\Delta_\eta\|_\infty = \sup_{(u,v)\in\Omega\times\Omega} |\Delta_\eta(u,v)|.$$

Given a measure μ on Ω, another choice is

$$\|\Delta_\eta\|_{L^2(\Omega\times\Omega)} = \left(\int_{\Omega\times\Omega} |\Delta_\eta(u,v)|^2 \, d(\mu\otimes\mu)(u,v)\right)^{1/2}.$$

However, a bijection may not exist. For the general case, consider Gromov–Hausdorff correspondences.

1.3.3 Gromov-Hausdorff distance

Recall (Hausdorff and Gromov–Hausdorff distances). For subsets A, B of a metric space Z, the Hausdorff distance is

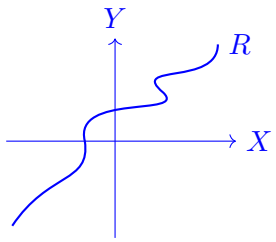
$$d_H^Z(A,B) = \inf\{\varepsilon > 0 : A \subseteq B_\varepsilon,\ B \subseteq A_\varepsilon\},$$

where A_ε is the ε-neighborhood of A. For compact metric spaces (X, d_X) and (Y, d_Y), define

$$d_{GH}(X,Y) = \inf_{Z,\varphi,\psi} d_H^Z(\varphi(X),\psi(Y)),$$

taking the infimum over metric spaces Z and isometric embeddings $\varphi : X \to Z$ and $\psi : Y \to Z$. See Burago, Burago, and Ivanov, Definition 7.3.10, p. 254.

Recall (Correspondences). Let X,Y be compact metric space. $R \subseteq X \times Y$ is a correspondence if the projections $\pi_X : R \to X$ and $\pi_Y : R \to Y$ are surjective. See Burago, Burago, and Ivanov, Definition 7.3.17, p. 256. For example, if $X = \{a, b\}$ and $Y = \{1, 2, 3\}$, then $R = \{(a,1),(a,2),(b,3)\}$ is a correspondence. The sketch below illustrates a relation between $X = Y = \mathbb{R}$.



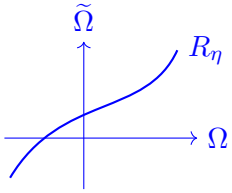
The distortion of a correspondence is

$$\operatorname{dis}(R) = \sup_{\substack{(x,y)\in R\\(x',y')\in R}} |d_X(x,x') - d_Y(y,y')|.$$

The correspondence formula (Theorem 7.3.25) gives

$$d_{GH}(X,Y) = \tfrac{1}{2} \inf_{R\in \mathrm{Corr}(X,Y)} \operatorname{dis}(R).$$

For a bijection $\eta : \Omega \to \widetilde{\Omega}$, its graph $R_\eta = \{(u,\eta(u)) : u \in \Omega\}$ is a correspondence because η is surjective.



For the supremum norm, this gives

$$\begin{aligned} d_{\mathcal{D}}(\Omega, \widetilde{\Omega}) &= \inf_{\eta \text{ bijective}} \sup_{u,v \in \Omega} |d(u,v) - \widetilde{d}(\eta(u), \eta(v))| \\ &= \inf_{\eta \text{ bijective}} \operatorname{dis}(R_\eta) \\ &\geq \inf_{R \in \operatorname{Corr}(\Omega, \widetilde{\Omega})} \operatorname{dis}(R) = 2 d_{GH}(\Omega, \widetilde{\Omega}). \end{aligned}$$

The inequality follows because $\{R_\eta : \eta \text{ bijective}\} \subseteq \operatorname{Corr}(\Omega, \widetilde{\Omega})$.

Definition 1.7 (Deformed signal)**.** *For a bijection η, define*

$$\widetilde{x} = x \circ \eta^{-1} \in \mathcal{X}(\widetilde{\Omega}), \qquad \widetilde{x}(\eta(u)) = x(u).$$

$\widetilde{x}$ is called the deformed signal.

Definition 1.8 (stable to domain deformation)**.** *Define $\mathcal{X}(\mathcal{D}) = \{(x, \Omega) : \Omega \in \mathcal{D},\ x \in \mathcal{X}(\Omega)\}$. We call $f : \mathcal{X}(\mathcal{D}) \to \mathcal{Y}$ is stable to domain deformation if*

$$\left\| f(x, \Omega) - f(\widetilde{x}, \widetilde{\Omega}) \right\| \leq C \, \|x\| \; d_{\mathcal{D}}(\Omega, \widetilde{\Omega})$$

1.4 Fourier Transform and Scale Separation

1.4.1 Fourier Representation and Continuous Wavelet Transform

Definition 1.9 (Fourier transform in one dimension))**.** *Let $\Omega = \mathbb{R}$. For $x \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, define*

$$\widehat{x}(\xi) = \int_{-\infty}^{\infty} x(u) e^{-i\xi u} \, du.$$

When $\widehat{x} \in L^1(\mathbb{R})$, the inverse formula gives, almost everywhere,

$$x(u) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{x}(\xi) e^{i\xi u} \, d\xi.$$

The extension to all of L^2 is described below.

Definition 1.10 (convolution theorem)**.** *The convolution of two functions is*

$$(x * \theta)(u) = \int_{-\infty}^{\infty} x(v) \theta(u - v) \, dv.$$

Convolution theorem If $x \in L^2$ and $\theta \in L^1$, then

$$\widehat{x * \theta} = \widehat{x} \, \widehat{\theta} \quad \text{in } L^2.$$

The right-hand side belongs to L^2 since $\widehat{\theta}$ is bounded.

Remark (Defining the transform on L^2). First define the linear Fourier transform $\mathcal{F} : L^1(\mathbb{R}) \to C^0(\mathbb{R})$ by

$$(\mathcal{F}f)(\xi) = \int_{\mathbb{R}} f(u)e^{-i\xi u}\,du.$$

For a general $f \in L^2$, the integral need not be absolutely convergent. We use the density of $L^1 \cap L^2$ and completeness of L^2(it is a Banach space) to extend $\mathcal{F}$. Plancherel gives

$$\boxed{\|\widehat{x}\|_2^2 = 2\pi \|x\|_2^2, \qquad x \in L^1 \cap L^2.}$$

Let $f_n \in L^1 \cap L^2$ converge to f in L^2. Then (f_n) is Cauchy in L^2, and

$$\left\|\widehat{f}_n - \widehat{f}_m\right\|_2 = \sqrt{2\pi}\,\|f_n - f_m\|_2\,.$$

Thus $(\widehat{f}_n)$ is Cauchy in L^2 and converges to some $g \in L^2$. The same estimate applied to two approximating sequences shows that the limit is independent of the choice. Taking $f_n = f\mathbf{1}_{[-n,n]}$(Then f_n is in $L^1 \cap L^2$) gives

$$\widehat{f} = \lim_{n\to\infty} \mathcal{F}f_n = \lim_{n\to\infty} \int_{-n}^{n} f(u)e^{-i(\,\cdot\,)u}\,du \qquad \text{in } L^2.$$

Definition 1.11 (Continuous wavelet transform)**.** *Let $\psi \in L^2(\mathbb{R})$ be a localized oscillatory function called the mother wavelet. Its wavelet atoms are*

$$\psi_{u,\xi}(v) = \xi^{-1/2}\psi\left(\frac{v-u}{\xi}\right), \qquad u \in \mathbb{R}, \quad \xi > 0.$$

Here u is the position and ξ is the spatial scale. The factor $\xi^{-1/2}$ gives $\|\psi_{u,\xi}\|_2^2 = \|\psi\|_2^2$. The CWT is

$$(W_\psi x)(u,\xi) = \xi^{-1/2} \int_{\mathbb{R}} x(v)\overline{\psi\left(\frac{v-u}{\xi}\right)}\,dv = \langle x, \psi_{u,\xi}\rangle_{L^2}.$$

It measures how close the signal x matches the wavelet atom $\psi_{u,\xi}$.

The difference between Fourier and CWT

1. The Fourier transform tells us which frequencies are present.
2. The CWT helps us understand where and at what scales oscillations occur.

One can sample the atoms dyadically, taking $\xi = 2^{-j}$ and $u = 2^{-j}k$ for integers j, k. Wavelets capture regularity properties such as piecewise smoothness.

1.4.2 Translation Invariance of Fourier and Deformation Stability of CWT

Shift property of the Fourier representation. Let $(S_v x)(u) = x(u - v)$. For $x \in L^1 \cap L^2$, the substitution $t = u - v$ gives

$$\begin{aligned}
\widehat{S_v x}(\xi) &= \int_{\mathbb{R}} x(u-v)e^{-i\xi u}\,du \\
&= \int_{\mathbb{R}} x(t)e^{-i\xi(t+v)}\,dt \\
&= e^{-i\xi v}\int_{\mathbb{R}} x(t)e^{-i\xi t}\,dt \\
&= e^{-i\xi v}\widehat{x}(\xi).
\end{aligned}$$

The identity extends to L^2 by density. In particular, $|\widehat{S_v x}| = |\widehat{x}|$, so the Fourier modulus is translation invariant.

Fourier is Sensitive to deformations Consider a deformation $\tau(u) = u - \widetilde{\tau}(u)$ and write

$$(\rho(\tau)x)(u) = x(u - \widetilde{\tau}(u)), \qquad \|\nabla\widetilde{\tau}\|_\infty = \sup_{u\in\Omega} \|\nabla\widetilde{\tau}(u)\| \le \varepsilon.$$

Even for small ε, the relative change

$$\frac{\|f(\rho(\tau)x) - f(x)\|}{\|x\|}$$

can be of order one uniformly over choices of signals and small deformations. Here $f(x) = |\widehat{x}|$. A slight zoom, for instance, can change the Fourier representation of an image.

Multiscale representation is deformation stable The wavelet representation is has the following approximate equivariance estimate under suitable assumptions on the wavelet and deformation:

$$\frac{\|\rho(\tau)(W_\psi x) - W_\psi(\rho(\tau)x)\|}{\|x\|} = O(\varepsilon).$$

On the wavelet output, $\rho(\tau)$ acts on the position variable, leaving the scale variable fixed.

$$\begin{array}{ccc} x & \xrightarrow{\quad\rho(\tau)\quad} & \rho(\tau)x \\ {\scriptstyle W_\psi}\downarrow & & \downarrow{\scriptstyle W_\psi} \\ W_\psi x & \xrightarrow[\rho(\tau)]{} & \rho(\tau)(W_\psi x) \approx W_\psi(\rho(\tau)x) \end{array}$$

which is also called approximately equivariant.

Definition 1.12 (Coarsening, Pooling, and Downsampling)**.** *Let Ω be a finite domain and let $\mathcal{X}(\Omega; V) = \{x : \Omega \to V\}$ be the space of V-valued signals on Ω.*

Coarsening. *In the partition-based setting, a coarsening of Ω is specified by a surjective map*

$$q : \Omega \longrightarrow \Omega_{\mathrm{c}}, \qquad |\Omega_{\mathrm{c}}| < |\Omega|.$$

Each coarse site $u \in \Omega_{\mathrm{c}}$ represents the block $B_u = q^{-1}(\{u\})$. These blocks form a partition of Ω. Equivalently, Ω_{c} can be identified with the quotient $\Omega/\!\sim$, where

$$v \sim w \quad \Longleftrightarrow \quad q(v) = q(w).$$

The map q specifies which fine sites belong to each coarse site. If the domain carries additional structure, such as a metric or graph edges, the construction must also specify what structure the coarse domain carries.

Pooling. *Pooling aggregates signal values over prescribed neighborhoods. For an output index set I, choose a nonempty neighborhood $N_u \subseteq \Omega$ and an aggregation rule $a_u : V^{N_u} \to W$ for each $u \in I$. The pooling operator is*

$$P : \mathcal{X}(\Omega; V) \longrightarrow \mathcal{X}(I; W), \qquad (Px)(u) = a_u(x|_{N_u}).$$

For block average pooling on a coarsening q, take $I = \Omega_c$ and $N_u = B_u$. For vector-valued signals,

$$(P_{\text{avg}}x)(u) = \frac{1}{|B_u|} \sum_{v \in B_u} x(v).$$

Pooling windows may overlap, and stride-one pooling can keep the original output resolution.

Downsampling. *Downsampling retains selected samples. Given an injective map $i : I \hookrightarrow \Omega$ with $|I| < |\Omega|$, define*

$$D_i : \mathcal{X}(\Omega; V) \longrightarrow \mathcal{X}(I; V), \qquad (D_i x)(u) = x(i(u)).$$

If each block B_u has a chosen representative $i(u)$, then $q \circ i = \mathrm{id}_{\Omega_c}$, and downsampling keeps one value per block.

Example. Let $\Omega = \{0, \dots, 2m-1\}$ and $\Omega_c = \{0, \dots, m-1\}$. Group adjacent sites in pairs and choose the even site as the representative of each pair. Then

$$\begin{aligned} q(j) &= \lfloor j/2 \rfloor && \text{(coarsening)}, \\ (P_{\text{avg}}x)(k) &= \tfrac{1}{2}(x(2k) + x(2k+1)) && \text{(average pooling)}, \\ (D_i x)(k) &= x(2k) && \text{(downsampling)}. \end{aligned}$$

The same average pooling operator can be written as $P_{\text{avg}} = D_i \circ A$, where $(Ax)(j) = \frac{1}{2}(x(j) + x(j+1))$ with periodic indexing. Here A computes averages on the original grid, and D_i retains the averages at even sites.

Let $P : \mathcal{X}(\Omega) \to \mathcal{X}(\Omega')$ be the coarsening map and let $f : \mathcal{X}(\Omega) \to \mathcal{Y}$ and $f' : \mathcal{X}(\Omega') \to \mathcal{Y}$ be classifiers. We want $f \approx f' \circ P$. Let $\mathcal{X}_j(\Omega_j, C_j)$ denote signals over the coarsened domain Ω_j. Informally, $f : \mathcal{X}(\Omega) \to \mathcal{Y}$ is locally stable at scale j if

$$f \approx f_j \circ P_j, \qquad P_j : \mathcal{X}(\Omega) \to \mathcal{X}_j(\Omega_j, C_j),$$

where P_j is a possibly nonlinear coarse-graining map. Long-range interactions in f on Ω can then be modeled by short-range interactions in f_j on Ω_j. Pooling in a CNN is an example.

1.5 Geometric Priors to Neural Network Architectures

Let G act on Ω and let $f : \mathcal{X}(\Omega, C) \to Z$ represent the data.

Symmetry The network satisfies invariance or equivariance under the group action.

Stability The change in the response to a small deformation is controlled.

Scale separation The network extracts information at different scales.

Assume $\mathcal{X}(\Omega, C)$ and Z are finite-dimensional normed vector spaces. Assume G is a locally compact Hausdorff topological group with Haar measure μ, where $0 < \mu(G) < \infty$, and that the group action is continuous.

1.5.1 Linear Invariant Maps

If f is G-invariant and linear, then it is continuous by finite dimensionality. Hence

$$\begin{aligned} f(x) &= \left(\frac{1}{\mu(G)}\int_G d\mu(g)\right) f(x) = \frac{1}{\mu(G)}\int_G f(x)\, d\mu(g) \\ &= \frac{1}{\mu(G)}\int_G f(g\cdot x)\, d\mu(g) \\ &= f\left(\frac{1}{\mu(G)}\int_G g\cdot x\, d\mu(g)\right). \end{aligned}$$

The third equality uses invariance. The last equality uses linearity and continuity, so f commutes with the vector-valued integral. Define the group averaging operator $A : \mathcal{X}(\Omega, C) \to \mathcal{X}(\Omega, C)$ by

$$Ax = \frac{1}{\mu(G)}\int_G g\cdot x\, d\mu(g).$$

Thus $f(x) = f(Ax)$, with the orbit map and f related by

$$G \xrightarrow{g\mapsto g\cdot x} \mathcal{X}(\Omega, C) \xrightarrow{f} Z.$$

Linear invariant maps have limited expressive power.

Example (Images on a periodic grid). Let $\Omega = \mathbb{Z}_n \times \mathbb{Z}_n$ and $x : \Omega \to \mathbb{R}^3$. For the translation group $G = \mathbb{Z}_n^2$, the action is $(g\cdot x)(u) = x(u-g)$. Using counting measure,

$$(Ax)(u) = \frac{1}{|G|}\sum_{g\in G} x(u-g) = \frac{1}{|G|}\sum_{v\in\Omega} x(v) = \frac{1}{n^2}\sum_{v\in\Omega} x(v).$$

Every pixel of Ax has the same color, the average color of x.

1.5.2 Linear Equivariant Maps and Pointwise Nonlinearities

Let $B : \mathcal{X}(\Omega, C) \to \mathcal{X}(\Omega, C')$ be linear and equivariant, and let $\sigma : C' \to C''$ be an arbitrary pointwise nonlinearity, acting on signals by $(\sigma x)(u) = \sigma(x(u))$. For the action $(g\cdot x)(u) = x(g^{-1}u)$ on the signal domain, the composition $U = \sigma \circ B : \mathcal{X}(\Omega, C) \to \mathcal{X}(\Omega, C'')$ is equivariant.

Proof.

$$\begin{aligned} U(g\cdot x)(u) &= \sigma\big(B(g\cdot x)(u)\big) = \sigma\big((g\cdot Bx)(u)\big) \\ &= \sigma\big((Bx)(g^{-1}u)\big) = (Ux)(g^{-1}u) = (g\cdot Ux)(u). \end{aligned}$$

□

Example (Group averaging after an equivariant map). If A averages the output signals and U is equivariant, then $A \circ U : \mathcal{X}(\Omega, C) \to \mathcal{X}(\Omega, C'')$ is G-invariant. Use normalized Haar measure $d\nu = d\mu/\mu(G)$, which is both left- and right-invariant for the compact group G. By right invariance,

$$A(g\cdot x) = \int_G h\cdot(g\cdot x)\, d\nu(h) = \int_G (hg)\cdot x\, d\nu(h) = \int_G k\cdot x\, d\nu(k) = Ax.$$

Consequently,

$$(A \circ U)(g \cdot x) = A(g \cdot Ux) = A(Ux) = (A \circ U)(x).$$

Moreover, using linearity of the group action and left invariance,

$$g \cdot (Ax) = \int_G g \cdot (h \cdot x)\, d\nu(h) = \int_G (gh) \cdot x\, d\nu(h) = Ax.$$

Thus the output of averaging is itself a G-fixed signal. Global invariance alone does not guarantee deformation stability.

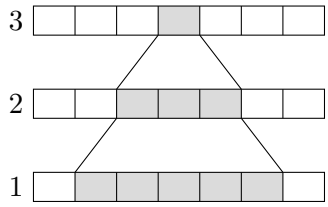
Definition (Localized equivariant map). Let Ω be a metric space. An equivariant map U is localized if $(Ux)(u)$ depends only on the values $x(v)$ for $v \in \mathcal{N}_u = \{v : d(u, v) \leq r\}$. The neighborhood $\mathcal{N}_u$ is the receptive field at u.

1.5.3 Multilayer Composition

A single localized equivariant map (LEM) only sees its receptive field. To capture long-range interactions, consider a composition

$$U_J \circ U_{J-1} \circ \cdots \circ U_1 : \mathcal{X}(\Omega, C) \to \mathcal{X}(\Omega, \widetilde{C}).$$

1. Composition can expand the receptive field. In the sketch, the central site in layer 3 depends on three sites in layer 2 and five sites in layer 1.



2. Composition preserves equivariance. Consider

$$\mathcal{X}(\Omega, C^0) \xrightarrow{U_1} \mathcal{X}(\Omega, C^1) \xrightarrow{U_2} \mathcal{X}(\Omega, C^2),$$

with induced representations ρ_k on $\mathcal{X}(\Omega, C^k)$. Equivariance means $U_1\rho_0(g) = \rho_1(g)U_1$ and $U_2\rho_1(g) = \rho_2(g)U_2$. Thus

$$(U_2 \circ U_1)\rho_0(g) = U_2\rho_1(g)U_1 = \rho_2(g)(U_2 \circ U_1).$$

1.5.4 The Blueprint of Geometric Deep Learning

The blueprint $(\Omega, G, B, \sigma, P, A)$ combines four building blocks: a local equivariant map, a global invariant map, a coarsening operator, and a nonlinear map. Let Ω' be a coarsened version of Ω (for example, a subset).

1. Previously $B : \mathcal{X}(\Omega, C) \to \mathcal{X}(\Omega, C')$ was equivariant. Now allow $B : \mathcal{X}(\Omega, C) \to \mathcal{X}(\Omega', C')$. Given a well-defined action of G on Ω' with induced representation ρ', equivariance means

$$B \circ \rho(g) = \rho'(g) \circ B.$$

2. We want local pooling $P : \mathcal{X}(\Omega', C'') \to \mathcal{X}(\Omega'', C'')$ to be equivariant

3. The final map A takes the last signal space to $\mathcal{Y}$ and is G-invariant.

With compatible actions and pointwise nonlinearities, the resulting map is G-invariant:

$$f = A \circ \sigma_J \circ B_J \circ (P_{J-1} \circ \sigma_{J-1} \circ B_{J-1}) \circ \cdots \circ (P_1 \circ \sigma_1 \circ B_1).$$

For example, the first block has types

$$\mathcal{X}(\Omega, C) \xrightarrow{B_1} \mathcal{X}(\Omega', C') \xrightarrow{\sigma_1} \mathcal{X}(\Omega', C'') \xrightarrow{P_1} \mathcal{X}(\Omega'', C'').$$

Questions. What about deformation stability? What is the relation between LEMs and stability?

Examples of Architectures, Domains, and Symmetries

Architecture	Domain	Symmetry
CNN	Grid	Translations
Spherical CNN	Sphere / SO(3)	Rotations SO(3)
Intrinsic / Mesh CNN	Manifold	Iso(Ω) / gauge symmetry (e.g. SO(2))
GNN	Graph	Permutations Σ_n
Deep Sets	Set	Permutations Σ_n
Transformer	Complete graph	Permutations Σ_n (without positional information)
LSTM	1D grid	Time warping (as a modeling prior)